A PROJECT REPORT

ON

FOUNDATION COURSE
(TINKERING & ELEMENTS OF DESIGN)

CLUBSPHERE

Submitted in partial fulfillment of the requirement for the award of Degree of
Bachelor of Technology in Computer Science & Engineering

Submitted to:

Rajasthan Technical University, Kota (Raj.)

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CERTIFICATE OF APPROVAL

The Project Report entitled "**ClubSphere**" submitted by **"Akanksha Sharma (24EARCS012), Avanshi Kothari (24EARCS031), Avantika Mishra (24EARCS032) Aditi Maheshwari (24EARCS007) "** has been examined by us and is hereby approved for carrying out the project leading to the award of degree "**Bachelor of Technology in Computer Science & Engineering"**. By this approval the undersigned does not necessarily endorse or approve any statement made, opinion expressed or conclusion drawn therein, but approve the pursuance of project only for the above mentioned purpose.

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This is to certify that the work embodied in this Project entitled "**ClubSphere**" being submitted by **"Akanksha Sharma (24EARCS012), Avanshi Kothari (24EARCS031), Avantika Mishra (24EARCS032) Aditi Maheshwari (24EARCS007) "** in partial fulfillment of the requirement for the award of "**Bachelor of Technology in Computer Science & Engineering"** to Rajasthan Technical University, Kota (Raj.) during the academic year 2026-27 is a record of bonafide piece of work, carried out by them under our supervision and guidance in the **"Department of Computer Science & Engineering", Arya College of Engineering & Information Technology, Jaipur**.

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DECLARATION

We "**Avanshi Kothari, Avantika Mishra, Aditi Maheshwari, Akanksha Sharma**", students of **Bachelor of Technology** in **Computer Science & Engineering, session 2025-26, Arya College of Engineering & Information Technology, Jaipur**, here by declare that the work presented in this Project entitled "**ClubSphere**" is the outcome of our own work, is bonafide and correct to the best of our knowledge and this work has been carried out taking care of Engineering Ethics. The work presented does not infringe any patented work and has not been submitted to any other University or anywhere else for the award of any degree or any professional diploma.

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Finally, we sincerely appreciate the collaborative efforts, dedication, and teamwork shown by all the members of our project group:

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ABSTRACT

Arya College ClubSphere is a proposed centralized web-based application for managing student clubs and college events. Colleges often have **multiple technical, cultural, sports, creative, and social clubs, each maintaining its own event announcements, membership lists, registration forms, and activity records. When this information is distributed across posters, messaging groups, spreadsheets, and paper records,** it can become difficult for students to discover opportunities and for administrators to monitor activity across the institution.

ClubSphere brings club information, user roles, event listings, event registration, QR-based registration access, task updates, notifications, and reports into one platform. The system is designed around four roles: **Admin/Professor, Club Head, Club Member, and Student**. The Admin can oversee all **26 clubs,** while Club Heads manage only their assigned clubs. Club Members can view club activities and can explore clubs and register for events.

The proposed technology stack includes assigned tasks, and Students **HTML5, CSS3, JavaScript, Bootstrap, Java, JSP, Servlets, JDBC, MySQL, and Apache Tomcat**. The Java web application structure separates the presentation layer, request handling, and database access. QR codes provide a convenient route from event posters to the relevant registration page. The project aims to reduce manual effort, improve the organization of club information, and provide a consistent experience for students and club administrators. **ClubSphere** is an academic software project. The final report should be aligned with the features actually implemented and tested in the submitted application; advanced capabilities such as automated notifications, payment processing, or analytics may require additional development.

Keywords: Java, JSP, Servlets, JDBC, MySQL, student clubs, event management, role-based access, QR registration. It simplifies the appointment process, improves user experience, and demonstrates how technology can solve real-world problems in the medical field.

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CHAPTER 1

INTRODUCTION

1.1 Background of the Study

Student clubs are an important part of college life. They help students develop technical skills, creativity, leadership, communication, teamwork, and professional confidence. A college may operate many clubs at the same time, with each club arranging workshops, competitions, cultural programmes, social initiatives, and other activities. When club information is managed separately, students may need to follow multiple social media groups or check different notice boards to learn about activities. Club coordinators may also have to maintain membership lists, event registrations, and attendance records manually. These processes can become difficult to coordinate when many clubs and events are active. Arya College ClubSphere is proposed as a single web-based platform for managing these activities. It provides a common location to discover clubs, publish event information, collect registrations, and organize club records.

1.2 Motivation of the Project

The motivation for ClubSphere is to make participation in college activities easier and to reduce repetitive administrative work. Students should be able to discover clubs and events without searching through scattered announcements. Club Heads should be able to manage their own activities in an organized manner, while the Admin should have a college-wide overview. A QR code linked to an event registration page can also make it easier for a student to register after seeing an event poster. The proposed Java-based architecture provides a structured way to handle user requests, apply role permissions, and store data in a relational database.

CHAPTER 2

PROBEM STATEMENT

2.1. Delay in Medical Consultation

- Managing multiple student clubs through disconnected tools can create information gaps and duplicated effort. Event details may be shared through posters or messaging groups, while registrations are collected through separate forms and membership records are maintained in spreadsheets. This arrangement may work for a small activity but becomes harder to monitor as the number of clubs and participants grows.

- **2.2. Lack of Accessibility**

- A centralized application is needed to maintain consistent club and event information, define clear responsibilities for different users, and make registration records easier to access. ClubSphere addresses this need by organizing the main club-management processes in one application.

2.3. Inefficient Appointment Management

- To design a role-based web application that centralizes the management of 26 college clubs, club members, events, event registrations, announcements, tasks, and reports, using Java web technologies and a MySQL database, with QR codes providing direct access to event registration pages.
- ClubSphere addresses this need by organizing the main club-management processes in one application.

2.4. Lack of Instant Medical Guidance

- In many cases, users require quick medical advice for minor issues, but:
- Doctors are not immediately available
- No platform exists for instant consultation
- Patients rely on unreliable sources for information
- This can lead to incorrect decisions and health risks.

2.5. Dependency on Physical Visits

- Most healthcare services require patients to physically visit hospitals, which leads to:
- Time consumption
- Travel expenses

2.6. Poor Data Management

- Existing systems often lack proper digital record management:
- Patient history is not easily accessible
- Data is stored manually or inefficiently
- Lack of data security and organization
- This affects diagnosis and treatment continuity.

2.7. Need for the Proposed System

- To overcome the above issues, there is a strong need for a modern digital solution that:
- Provides instant access to healthcare services
- Enables online consultation and appointment booking
- Ensures efficient data management

- Offers reliable and user-friendly interface
- The application is proposed as a solution to meet these requirements and improve the overall healthcare experience.

2.8. Proposed Solution

- The proposed solution is the development of the Quick Qure Application, a mobile-based healthcare platform that integrates multiple services into a single system.
- Key Features of the Solution:
- Online doctor consultation
- Digital appointment booking system
- User-friendly mobile interface
- Secure data storage and management
- Quick access to medical information
- This solution aims to simplify healthcare access while maintaining efficiency and reliability.

2.9. Objectives Derived from the Problem

- Based on the identified problems, the project aims to:
- Eliminate delays in medical consultation
- Improve accessibility of healthcare services
- Develop an efficient appointment management system
- Provide instant medical guidance
- Create a scalable and user-friendly application

CHAPTER 3

OBJECTIVES OF THE PROJECT

3.1. Primary Objective

The primary objective of **Arya College ClubSphere** is to develop a centralized web-based platform that simplifies and streamlines the management of college clubs and events. The system aims to provide a common platform where students can explore clubs, access event information, register for activities using QR codes, and stay updated on college events. It also enables administrators and club heads to manage club activities, event registrations, and student participation efficiently. By reducing manual paperwork and improving communication, ClubSphere aims to make college club and event management more organized, efficient, and user-friendly.

3.2. Specific Objectives of the Project

- To provide a centralized platform for managing college clubs and events efficiently.
- To simplify event registration using QR codes and maintain student registration records.
- To provide role-based access for administrators, professors, club heads, and students.
- To reduce manual paperwork, improve communication, and encourage student participation in extracurricular activities.

3.3 Implementation of Online Doctor Consultation

One of the key objectives is to enable users to consult doctors remotely. This feature allows patients to:

- Seek medical advice from anywhere
- Avoid unnecessary hospital visits
- Save time and effort
- This improves convenience and accessibility.

3.4 Efficient Appointment Booking System

- The project provides practical experience in requirement analysis, role-based design, web interfaces, server-side request handling, database connectivity, relational data modelling, form validation, testing, and collaborative software development

3.5. Improvement of Healthcare Accessibility

- The application is designed to make healthcare services accessible to a wider audience, including people in remote or rural areas where medical facilities are limited.

3.6 Development of a User-Friendly Interface

- The project aims to develop a simple, attractive, and user-friendly interface for students and club administrators. It provides easy navigation to explore clubs, view upcoming events, and register for activities. The interface is designed to be responsive and accessible across different devices. This helps users manage club activities efficiently with minimal effort.

3.7.Secure Data Management

The project aims to ensure that all user data, including personal and medical Information , is stored securely. This includes:

- Safe storage of patient records
- Protection of user privacy
- Reliable database management

3.8. Reduction of Hospital Workload

By providing online services, the application helps in reducing the burden on hospitals by:

- Decreasing unnecessary visits

- Managing patient flow efficiently
- Improving resource utilization

- **3.9.Providing Instant Medical Guidance**

- The system aims to provide users with quick access to basic medical information and guidance, helping them make informed decisions regarding their health.

3.10. Development of a Cost-Effective Solution

- The application is designed to be economical and accessible, ensuring that users can benefit from healthcare services without high costs.

3.11 Scalability and Future Expansion

Another important objective is to design the system in a way that allows future enhancements, such as:

- Integration with AI-based diagnosis
- Video consultation features

- IoT-based health monitoring

CHAPTER 4

SCOPE OF THE PROJECT

- The scope of ClubSphere is to provide a centralized platform for managing college clubs and events. It allows students to explore clubs, view upcoming events, and register for activities using QR codes. Club administrators and club heads can manage club information, events, and student registrations efficiently. In the future, the system can be enhanced with mobile application support, online notifications, and attendance tracking.

4.1. Functional Scope

- The functional scope of ClubSphere includes user registration, login, and role-based access for different users. It allows students to explore college clubs, view events, and register using QR codes. Club heads and administrators can manage club details, events, and student registrations. The system maintains organized records to ensure efficient club and event management.

4.2. User Scope

- Students use the platform to explore clubs and register for events. Club Members view relevant club information, tasks, announcements, and activities. Club Heads manage their assigned club's information and events. Admin/Professor users oversee the overall system and can access college-wide information

4.3. Technical Scope

- The intended web stack includes HTML5, CSS3, JavaScript, Bootstrap, Java, JSP, Servlets, JDBC, MySQL, and Apache Tomcat. The user interface runs in a

modern browser. Java server-side components process requests and use JDBC to communicate with the MySQL database.

4.4. User Registration and Profile Management

- ClubSphere allows users to register and log in to the platform using their required credentials. Users can manage their profiles and access features according to their assigned roles, such as administrator, club head, club member, or student. The system maintains user information securely and provides authorized access to club activities and event registration. This feature ensures smooth user management and a personalized experience.

4.8. Operational Scope

The operational scope describes how the system will function in real-world scenarios.

- Users can access healthcare services from anywhere
- Doctors can manage appointments efficiently
- The system reduces manual work and improves workflow
- Provides a smooth interaction between users and healthcare providers

4.9. Academic Scope

- The academic scope of ClubSphere is to encourage students to participate in extracurricular activities alongside their studies. It provides information about college clubs, workshops, seminars, competitions, and other academic events. Students can explore opportunities to develop their technical skills,

creativity, teamwork, and leadership abilities. The platform supports the overall academic and personal development of students.

CHAPTER 5

Existing System

5.1 Description of the Existing Approach

In a manual or partly digital approach, club information is often communicated through notice boards, posters, messaging groups, social media posts, and

separate online forms. Member lists and attendance details may be stored in spreadsheets or paper files. Each club may follow its own process, so there may not be a consistent view of activities across the institution.

In many cases, healthcare services are not easily accessible, and the process of consulting doctors or booking appointments is time-consuming and inefficient. This section analyzes the working, advantages, and limitations of the current system to highlight the need for improvement.

5.2 Typical Workflow

Club prepares poster → Announcement is shared → Student finds the form → Student submits details → Club checks registrations manually → Participation records are updated separately

5.3.Advantages of the Existing Approach

- Simple to begin using without building a dedicated application.
- Messaging platforms can distribute information quickly.
- Paper records may be useful when internet access is unavailable.
- Separate forms can be sufficient for a small event with few participants.
- Receives prescription or treatment advice
- In case of appointment booking

5.3. Advantages of Existing System

Despite its limitations, the traditional system has some advantages:

- Direct face-to-face interaction with doctors
- Suitable for serious and emergency cases

- No dependency on internet or technology
- Established and widely accepted system

5.4. Limitations of Existing System

- Information is spread across several channels.
- Records may be duplicated or become outdated.
- A student may miss an event announcement.
- Club Heads may need to check multiple files and forms.
- College-wide summaries can require manual effort.
- Access permissions are difficult to manage consistently across separate tools
- Poor Record Management
- High Dependency on Physical Presence
- Lack of Digital Integration
- Increased Workload on Hospitals

5.5. Structure of the Existing System

- In the existing system, college club activities and event management are often handled manually using paper records, spreadsheets, and separate communication platforms. Students may need to contact club coordinators to obtain information about clubs and upcoming events. Event registration and record maintenance can be

time-consuming and difficult to manage. The absence of a centralized platform creates challenges in communication, data management, and student participation.

CHAPTER 6

PROPOSED SYSTEM

6.1 Explanation

- Arya College ClubSphere is proposed as a centralized web application that brings together club discovery, event information, registrations, and activity records. Instead of using a separate process for every club, users work through a common application with permissions determined by their role.

6.2 Main Features

- Central club directory: lists the 26 clubs and presents their profiles and activities.
- Role-specific dashboards: displays relevant actions and information for each type of user.
- Event management: allows authorized Club Heads to publish and maintain event information.
- Event registration: collects participant information and records registration status.
- QR-based entry: provides a convenient route from a poster or announcement to an event registration page.
- Member and task management: supports club membership visibility and task updates.
- Reports: helps authorized users review club, event, member, and registration information.

6.4 Benefits of the Proposed Design

- A common data structure improves consistency, role-based access clarifies responsibilities, and event-linked QR codes reduce the number of steps between discovering an event and opening its registration form. Centralized records can also make future reporting and analysis easier.

CHAPTER 7
SYSTEM REQUIREMENTS

7.1 Introduction

- System requirements describe the hardware, software, and functional capabilities needed to develop and use the application. The following specifications are suggested for a student development environment and may vary with deployment conditions.

- .**7.2 Functional Requirements •**
- The system should allow authorized users to sign in and sign out.
- The system should identify the user's role and show the relevant dashboard.
- Admin users should be able to manage or monitor all configured clubs.
- Club Heads should be able to manage events for their assigned clubs
- Students should be able to browse available clubs and events.
- Eligible users should be able to submit event registrations.
- The system should store registration details and make them available to authorized users.
- An event of QR code or QR link should open the appropriate registration page.
- The system should support viewing relevant announcements and task details.
- The system should provide summaries or reports where those modules are implemented.
- **7.3 Compatibility Note**
- Tomcat 10.1 uses the Jakarta Servlet namespace.
- Java web source code and dependencies should use compatible Jakarta APIs rather than older javax.servlet imports.
- The selected JDK version should also be compatible with the installed Tomcat release and any project libraries.

CHAPTER 8

ADVANTAGES OF THE SYSTEM

8.1 Centralized Information

- Club profiles, events, and related records can be accessed from one application rather than requiring students and coordinators to search through multiple sources.

8.2 Easier Event Discovery

- Students can explore the club directory and upcoming events, helping them find activities that match their interests.

8.3 Role-Based Control

- The four-role structure separates college-wide administration from club-level management and student participation. This reduces the risk of users accessing functions outside their responsibilities when the controls are correctly implemented.

8.4 QR-Based Convenience

- An event QR code can be printed on posters or shared digitally. Scanning it opens the event registration page and can reduce manual searching for the form.

8.5 Better Record Keeping

- Centralized records can help authorized users review membership, events, registrations, and participation. Reports may reduce repetitive compilation work.

8.6 Extensibility

- The Java web application and relational database provide a foundation for future additions such as notifications, payment integration, attendance tracking, certificate generation, and analytics.

CHAPTER 9

MODULES OF THE PROJECT

9.1 Authentication and User Management

- This module handles user login, logout, account details, role identification, and access control. After authentication, the application directs the user to the relevant dashboard. Role validation must be performed on the server as well as reflected in the user interface.

9.2 Club Management

- This module maintains information about the 26 college clubs, including club name, description, logo, assigned Club Head, status, and member list. Admin users can oversee the complete directory. Club-level users should only be able to modify records within their assigned permissions.

9.3 Event Management

- Authorized Club Heads can create event records with a title, description, date, time, venue, registration deadline, capacity, event type, and status. Editing and cancellation should preserve the accuracy of existing registration records.

9.4 Event Registration

- Students select an event and provide the required participant details. The application validates the form and records the registration in the database. Where group registration is enabled, the system can store a group name, group size, and participant details. Capacity and deadline checks should be performed before confirming registration.

9.5 QR-Based Registration

- A QR code represents a link to the registration page for a particular event. The user scans the code, opens the page, reviews event information, completes the form, and submits it. The server validates the submission and stores the registration. Create Event → Generate Event Link / QR → Publish Poster → Student Scans → Registration Page → Validate Form → Save Registration → Confirmation

9.6 Member and Task Management

- This module supports club membership visibility and the assignment of tasks to club members. A task may include a title, description, assignee, due date, and status such as Pending, In Progress, or Completed. Access to create or update tasks should follow the permissions of the assigned role.

9.7 Dashboard and Analytics

- The Admin dashboard may summarize the number of clubs, users, events, registrations, upcoming events, and recent activities. Club Head dashboards focus on their assigned club. Member and Student dashboards should remain focused on their own relevant activities and registrations. Counts should be calculated from actual database records rather than hard-coded values.

9.8 Notifications and Reports

- Notifications can be used for registration confirmations, new events, deadlines, announcements, and task updates. Reports may summarize events, registrations, club members, and participation. Email or SMS delivery should be treated as an additional integration unless it has been implemented and tested.

CHAPTER 10

SYSTEM DESIGN

10.1 Introduction

- System design describes how the user interface, Java request-handling components, database-access layer, and MySQL database work together. A layered design makes the application easier to test, maintain, and extend.

10.2 Proposed Architecture

- USER / BROWSER HTML5 + CSS3 + JavaScript + Bootstrap ↓ HTTP Request / HTTP Response ↓ JSP VIEW JAVA SERVLETS / CONTROLLERS ↓ JDBC / Prepared

Statements MYSQL DATABASE The browser presents forms and pages.

- JSP is used for dynamic views, while Servlets handle incoming requests, validate data, apply business rules, and call the data-access layer.
- JDBC connects the Java application to MySQL. Apache Tomcat hosts the web application.

10.3 Main System Components

- User interface: pages for login, dashboards, clubs, events, registration, tasks, and reports.
- Authentication and authorization: checks user credentials and role permissions.
- Application logic: validates requests and enforces rules such as event capacity and deadlines.
- Data access: uses JDBC and prepared statements to execute database operations.
- Database: stores users, clubs, memberships, events, registrations, tasks, and notifications.

- **10.4 High-Level Data Flow**

- User submits form → Server validates request → Application checks permissions and business rules → JDBC executes database operation → Server returns result → Browser displays confirmation or error

- **10.5 Input Design**

- Input forms should use clear labels, required-field indicators, sensible defaults, and validation messages.
- Common inputs include user name, email, role, club, event title, date/time, venue, registration type, group size, and participant details. Server-side validation is required even when browser-side validation is used.

10.6 Output Design

- The application should show readable event details, registration confirmation or failure, registration history, club membership information, task status, and relevant dashboard summaries. Error messages should help users correct the issue without exposing internal database or server details.

10.7 Database Design

- Overview The database should use primary keys to identify records and foreign keys to maintain relationships. A user can have a role and may be associated with a club through a membership record. A club can have multiple events, and an event can have multiple registrations. Group registration may require a separate participant table. Page 19ARYA COLLEGE CLUBSPHERE

10.8 Security Design

- • Use secure password storage rather than plain-text passwords.
- Use server-side role checks on every protected operation.
- Use JDBC PreparedStatement for user-supplied values.
- Validate and normalize input before storing it.
- Manage sessions carefully and invalidate them on logout.
- Avoid displaying stack traces, credentials, or database details to users.
- Restrict access to personal registration information to authorized users.

10.9 Suggested Entity

- Relationships
- USERS 1 CLUBS 1 * CLUB_MEMBERS *
- * EVENTS 1 REGISTRATIONS 1 *1 CLUBS
- * REGISTRATIONS * EVENT_PARTICIPANTS
- CLUBS 1 * TASKS; USERS 1 * NOTIFICATIONS
- These relationships are a conceptual starting point.
- The final ER diagram and table structure should reflect the actual implementation and constraints

CHAPTER 11

HOW THE SYSTEM WORKS

11.1 Introduction

- This chapter explains the typical user journey through ClubSphere, from login and club discovery to event registration and administrative review.

11.2 Step-by-Step Workflow

- Step 1 — User Login The user opens the application and enters their credentials. The server validates the credentials and creates a session if authentication succeeds.
- Step 2 — Role Identification The application retrieves the user's assigned role and displays the appropriate dashboard. Protected operations are checked again on the server when a request is submitted.
- Step 3 — Explore Clubs or Open Club Dashboard Students can browse the 26

clubs and review descriptions and activities. Club Heads and Club Members see the information and functions allowed for their assigned club.

- Step 4 — Event Creation An authorized Club Head enters event details such as title, date, venue, registration deadline, capacity, and event type. The server validates the details and stores the event record.

- Step 5 — QR Link Publication A registration URL is associated with the event. A QR code can be generated from that URL and placed on an event poster or shared digitally.

- Step 6 — Student Registration The student scans the QR code or opens the event from the event list, reviews the details, fills in the registration form, and submits it.

- Step 7 — Validation and Database Storage The application checks required fields, registration deadline, event capacity, and any duplicate-registration rule configured for the event. If valid, the registration is saved in MySQL.

- Step 8 — Confirmation and Review The application displays the result to the student. Authorized Club Heads and Admin users can view the stored registration according to their permissions. Email or SMS confirmation is available only if the corresponding integration is configured.

- **11.3 Overall Working Flow**

- Login → Role-Based Dashboard → Explore Clubs / Manage Club → View or Create Event → Open Registration Page → Submit Form → Validate → Save in MySQL → Confirmation → Review Records

- **11.4 Exception Handling**

- • Invalid login: show a general authentication error and allow another attempt.

- Unauthorized action: deny access and redirect or show an access-denied message.

- Missing or invalid form fields: display validation errors.

- Closed or full event: prevent new registrations and explain the reason.

- Database error: show a safe user-facing message and log technical details securely.

- Duplicate registration: follow the configured rule and avoid unintended duplicate records.

-